This page is part of the HL7 FHIR Implementation Guide: minimal Common Oncology Data Elements (mCODE) Release 1 - US Realm | STU1 (v3.0.0: STU (https://confluence.hl7.org/display/HL7/HL7+Balloting) 3) based on FHIR (HL7® FHIR® Standard) R4 (http://hl7.org/fhir/R4). The current version which supersedes this version is 4.0.0 (http://hl7.org/fhir/us/mcode). For a full list of available versions, see the Directory of published versions (http://hl7.org/fhir/us/mcode/history.html)

1 Home

Official URL: **`http://hl7.org/fhir/us/mcode/ImplementationGuide/hl7.fhir.us.mcode`**	*Version*: **`3.0.0`**
Draft as of 2023-10-25	*Computable Name*: `MinimalCommonOncologyDataElements`

1.1 Background

Cancer is among the leading causes of death worldwide. According to the National Cancer Institute, in the United States, 39.5 percent of men and women will be diagnosed with cancer at some point during their lifetimes. In 2020, an estimated 1,806,590 new cases of cancer will be diagnosed in the United States and 606,520 people will die from the disease. While these numbers are staggering, the silver lining in the wide prevalence of cancer is the potential to learn from treatment of millions of patients. If we had research-quality data from all cancer patients, it would enable better health outcomes. Today, we lack the data models, technologies, and methods to capture that data.

mCODE™ (short for Minimal Common Oncology Data Elements) is an initiative intended to increase interoperability by assembling a core set of structured data elements for oncology electronic health records (EHRs). mCODE is a step towards capturing research-quality data from the treatment of all cancer patients. This would enable the treatment of every cancer patient to contribute to comparative effectiveness analysis (CEA) of cancer treatments by allowing for easier methods of data exchange between health systems.

1.2 Overview

mCODE consists of approximately 30 FHIR profiles organized into six thematic groups. Groups are introduced for pedagogical purposes only and have no other meaning or consequence:

- Patient Information Group (group-patient.html)

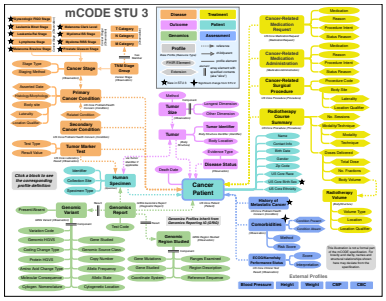
- Disease Characterization Group (group-disease.html)
- Health Assessment Group (group-assessment.html)
- Genomics Group (group-genomics.html)
- Cancer Treatments Group (group-treatment.html)
- Outcomes Group (group-outcome.html)

The overall scope of mCODE and the relationships between mCODE profiles is shown in the following diagram (not all details are shown). Click on items to see their underlying definitions. When viewing the diagram, please note:

1. mCODE does not define a specific set of information that must be collected for each cancer patient. Instead, mCODE presents a set of patterns or templates, including controlled terminologies, to be used when certain data in the above groups are shared. ***Example:*** *mCODE includes a FHIR profile for ECOG performance status, but not every cancer patient undergoes such an evaluation. However, if a patient DOES have an ECOG performance evaluation, that information must conform to mCODE's ECOG profile when shared.*

2. If mCODE does not have a profile for a certain type of data, that data still can be included with mCODE data using an appropriate resource or profile. The lack of an mCODE profile only means that mCODE does not impose requirements above and beyond the base FHIR specifications or US Core. ***Example:*** *mCODE does not include a profile for care teams. However, users can share care team information using FHIR's CareTeam resource, using a profile such as US Core CareTeam.*

3. Additional profiles can be derived from mCODE profiles. It is expected that mCODE will serve as the basis for future implementation guides dealing with new use cases, particular types of cancer, or specific demographic groups using mCODE as a base. ***Example:*** *Radiation oncologists and vendors collaborated on a radiation therapy implementation guide that extends mCODE and facilitates structured data sharing between radiation oncology information systems and other health information systems.*


mCODE STU 3
Disease
Treatment
Outcome
Patient
Genomics
Assessment
Profile
Cancer Stage
TNM Stage Group
Primary Cancer Condition
Secondary Cancer Condition
Tumor Marker Test
Human Specimen
Genomic Variant
Genomics Report
Tumor Size
Tumor
Cancer Patient
Disease Status
Cancer-Related Medication Request
Cancer-Related Medication Administration
Cancer-Related Surgical Procedure
Radiotherapy Course Summary
History of Metastatic Cancer
Comorbidities
ECOG/Karnofsky Performance Status
Radiotherapy Volume
External Profiles
Blood Pressure
Weight
Height
CAP
CRC

1.3 Data Dictionary

Readers should take note of the Data Dictionary (dictionary.html), a flattened list of data elements in mCODE in Microsoft Excel format. There is also a Data Dictionary Differential (dictionary.html#data-dictionary-differential) that compares STU 3 with STU 2 on an element-by-element basis.

The Data Dictionary comes with several caveats:

- Only must-support (MS) data elements are included. These are elements that implementers could be expected to provide meaningful support for (see Profile Conformance (conformance-profiles.html) for details).
- Profiles defined externally to mCODE, such as vital signs defined in base FHIR or US Core, are not included.
- Sub-elements of complex types such as CodeableConcept and Period are not explicitly included.

The actual number of data elements in a resource can vary widely. For example, there are 16 MS data elements in the CancerPatient (StructureDefinition-mcode-cancer-patient.html) profile, but the full resource defines more than 60 elements. Of these, only four are required (conformance-profiles.html#definition-of-required) (namely `identifier.system`, `identifier.value`, `name.family` or `name.given`, and `gender`). Moreover, different systems can implement different subsets of profiles, depending on their role in information workflows (see Profile Conformance (conformance-profiles.html) for details). Therefore, not every mCODE-compliant system will support the same set of data elements. However, whatever portion of this specification is implemented will be compatible with other systems, inasmuch as the implementations overlap.

1.4 Development History

In late 2018, American Society of Clinical Oncology (ASCO®), with the assistance of MITRE Corporation, convened a committee of twenty leading clinical experts in oncology, radiology, surgery, and public health. This group developed two use cases that drove the initial clinical data requirements for mCODE:

- **Use Case 1**: Comparative Effectiveness Analysis and Cooperative Decision Making (mCODE-UseCase-RCC.docx)
- **Use Case 2**: Comparative Effectiveness Analysis with Next Generation Sequencing (NGS) (mCODE-UseCase-NGS.docx)

After the initial analysis, an open survey was conducted to validate and prioritize the data elements from these use cases. Down-scoping was conducted based on the likelihood the data elements would be found in current EHRs, and if collecting the data would place undue burden on clinicians. In 2019, mCODE was balloted and approved as an HL7 Standard for Trial Use (STU 1). In the ensuing period, mCODE was piloted at a number of clinical sites, facilitated by the CodeX FHIR Accelerator. Use cases are tracked here. Incorporating that experience, the second trial use publication (STU 2) was balloted in May 2021, and published in January 2022. A non-balloted update containing terminology updates, version 2.1, was published in March 2023. STU 3 was balloted in April-May 2023.

1.5 Sources

In addition to information obtained from subject matter experts, several existing standards, nomenclatures, and guidelines were consulted during the development of this specification. These include:

- American Association of Physicists in Medicine (AAPM): Task Group 263 Final Report
- American College of Surgeons (ACS): Standards for Registry Entry (STORE)
- American Society for Radiation Oncology (ASTRO): Minimum Data Elements for Radiation Oncology
- American Society for Radiation Oncology (ASTRO) Commission on Cancer Workgroup: A Multidisciplinary Consensus Recommendation on a Synoptic Radiation Treatment Summary
- American Joint Committee on Cancer (AJCC): Staging Manual (8th Edition)
- Clinical Data Interchange Standards Consortium (CDISC): Therapeutic Area User Guides (TAUG)
- College of American Pathologists (CAP): Cancer Protocols
- The DICOM Standard
- HL7 CDA R2 IG: Reporting to Public Health Cancer Registries from Ambulatory Healthcare Providers
- HL7 FHIR Genomics Reporting Implementation Guidance (STU1)
- Human Genome Organization (HUGO): Gene Nomenclature Committee
- Human Genome Variation Society: Nomenclature for defining genomic variants.
- National Center for Biotechnology Genomic Test Registry (GTR)
- National Comprehensive Cancer Network (NCCN) Clinical Practice Guidelines in Oncology
- North American Association of Central Cancer Registries (NAACCR): 2018 Site-Specific Data Items Manual
- North American Association of Central Cancer Registries (NAACCR): Standards for Cancer Registries
- RECIST Guidelines v1.1

In addition, material was drawn from the US Core Implementation Guide and the Genomics Reporting Implementation Guide.

1.6 Understanding this Guide

mCODE is a “Domain of Knowledge” implementation guide (IG). The purpose is to show how to represent clinical concepts generally, not to have a complete set of agreements for interoperable exchanges.

The page layouts and symbols are explained in the FHIR documentation. In viewing a profile page, note that there are multiple views. The “Differential Table” view represents the difference between the current profile and its base resource or profile. When interpreting this view, bear in mind that the immediate parent may not be a base FHIR resource, but could be a US Core profile or another profile in this guide. The “Snapshot Table” represents the entire profile, with all elements.

In the event there are differences between the page renderings in this IG and the associated FHIR artifacts, the FHIR artifacts should be taken as the source of truth. In the event that an artifact’s snapshot is inconsistent with its differential, the differential should be taken as the source of truth.

1.7 Credits

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1.8 Contact Information

If you have questions or comments about this guide, you can join the conversation on chat.fhir.org, create an issue in the HL7 Jira, or reach out to any of the following individuals:

Topic	Who	Role	Email
Implementation and Use Cases	Dr. Su Chen	CodeX Accelerator Program Manager	suchen@mitre.org
CodeX Leadership	Dr. Travis Osterman	Chair, mCODE Executive Committee	travis.osterman@vumc.org
Oncology Domain Content	Dr. Sanjay Aneja	Chair, mCODE Technical Review Group	sanjay.aneja@yale.edu
Modeling and FHIR IG	Mark Kramer	Modeling Lead	mkramer@mitre.org

mCODE is an open source project and welcomes all contributors. The source code for this IG is maintained in the HL7 Github. All of the profiling work is done using FHIR Shorthand and SUSHI.

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